

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed **end-to-end platform architectures** for robotics systems, integrating sensing, compute, connectivity, and motion to create reusable building blocks, reducing integration time for new concepts by **30%**.
- Designed and executed **characterization and test automation** for robotics subsystems, generating performance and robustness data that informed key architecture choices and accelerated technology readiness.
- Built and iterated on **rapid prototypes** for advanced robotics concepts, encompassing electrical, mechanical, firmware, and software components to de-risk system feasibility.
- Authored **design documentation and integration guides** to transfer learnings and recommended system patterns to product teams, capturing critical tradeoffs and pitfalls.
- Managed **production support and incident response** for enterprise Java applications, resolving P1/P2 incidents within strict SLAs using tools like **Splunk** and **Dynatrace**.
- Automated data pipelines and reporting tasks using **Python scripting**, improving data processing efficiency for scheduled reports by **40%**.

SKILLS

Robotics Software & Controls – ROS/ROS2, Scripting, Data Analysis, Embedded Firmware, Real-time Development, Control Loops, Drivers, Timing, Interfaces

System Integration & Connectivity – System Integration, Connectivity, Telemetry, Industrial Interfaces, Networking, Edge Compute Integration

Hardware & Prototyping – Rapid Prototyping, Electrical Design, Mechanical Design, Firmware Development, Lab Measurement, Hardware Bring-up, Sensing, Power, Motion Subsystems

Software Development – Java, Spring Boot, Python, REST APIs, Docker, Linux, Git, Jenkins, CI/CD

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch)

Data & Analytics – SQL (PostgreSQL, MySQL, Oracle), Power BI, DAX, Power Query, Data Modelling, Data Warehousing, Tableau, R, pandas, NumPy, scikit-learn

Project Management & Support – Agile/Scrum, Jira, Confluence, Change Management, UAT Support, Incident Response

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Robotics Motion Control System Integration

- Developed a **real-time control loop** for a robotic arm actuator, integrating sensor feedback and command inputs using **Python** and **Linux**.
- Designed and implemented **system integration** for a new motion subsystem, connecting motor controllers, encoders, and a central compute unit via industrial interfaces.
- Generated **performance benchmarking data** for the motion control system, achieving a **15% improvement** in positional accuracy through firmware adjustments.

Automated Robotics Test Platform

- Built an **automated test framework** using **Python** to execute characterization and regression tests on robotic hardware prototypes.
- Integrated **Docker containers** for consistent test environment deployment, enabling repeatable execution across development and staging.
- Developed **dashboard visualizations** using **Power BI** to present test results, identifying critical failure modes and reducing debugging cycles by **25%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)